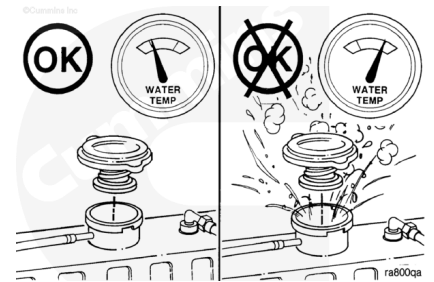


Trainings ▾

Drain

⚠ WARNING ⚠

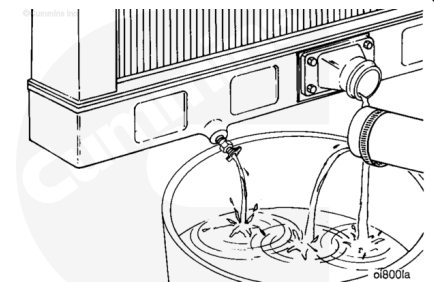
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Remove the radiator cap after the engine is cool.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

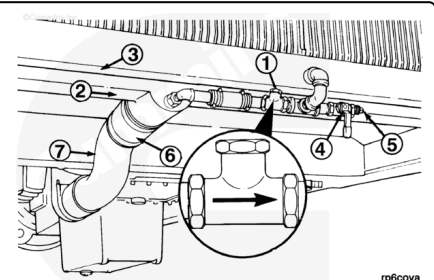


Position the vehicle on level ground. Open the draincock at the bottom of the radiator. Remove the lower radiator hose. Drain the cooling system.

LTA

⚠ WARNING ⚠

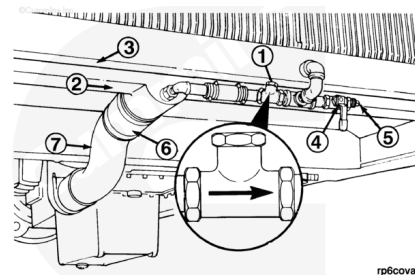
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Position the vehicle on level ground. Drain the coolant.

The low temperature aftercooler (LTA) cooling system is drained by removing a pipe plug (5) and opening a swing valve (4) located in the coolant plumbing connecting the bottom tank of the main engine and LTA radiator cores.

Note : On some radiators, the plumbing that connects the bottom tank of the engine and the low temperature aftercooler (LTA) radiator is incorporated internally. With this design, there will be two drain locations on the radiator bottom tank.



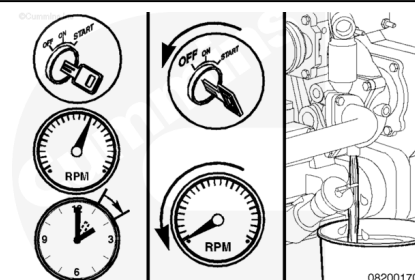
The plumbing between the radiator cores is designed to **only** allow coolant to flow from the main engine core (2) to the LTA core (3) during the fill procedures. To prevent coolant flow in the opposite direction during engine operation, a swing valve (1) **must** be included in the plumbing. The arrow on the swing valve **must** point toward the LTA core.

Draining the cooling system at the main engine radiator core by removing the hose (6) or the pipe (7) will **not** allow the LTA core to drain due to the swing valve.

Marine Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

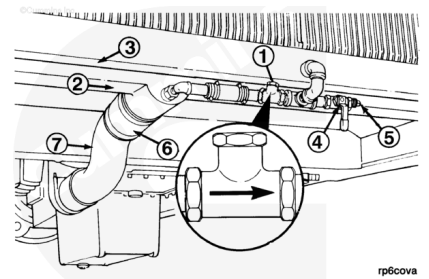
Operate the engine at normal operating temperatures, at least 85°C [185°F], for 1 to 1 ½ hours.

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

Flush

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Note : The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning time. Restore™ can be safely used up to twice the recommended concentration levels. Extremely scaled or fouled systems can require more than one cleaning.

Drain the cooling system.

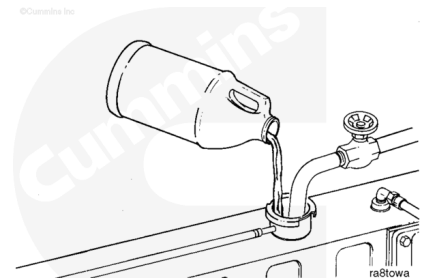
Do **not** remove the coolant filters.

⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

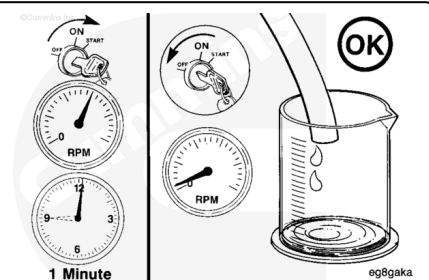
Immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, or equivalent, for each 38 to 57 liters [10 gal] of cooling system capacity, and fill the system with plain water.

Turn the heater temperature switch to “HIGH” to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

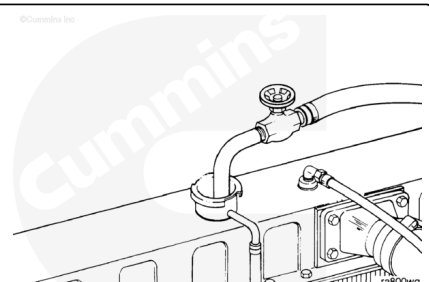


Operate the engine at a coolant temperature of at least 85°C [185°F] for 1 to 1-½ hours.

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.



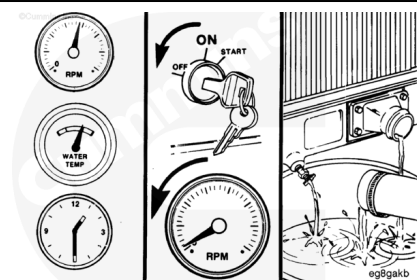
Fill the cooling system with clean water.



Operate the engine at high idle for 5 minutes with the coolant temperature above 85°C [185°F].

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

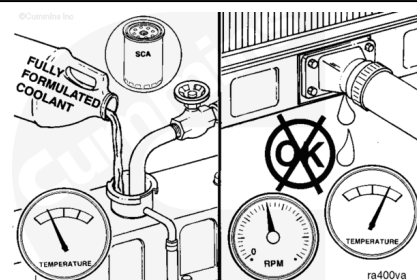
If the water drained is still dirty, the system **must** be flushed again until the water is clean.



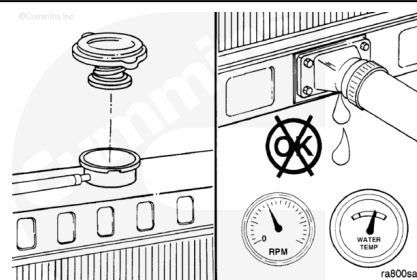
Install a new coolant filter and fill the cooling system with fully formulated coolant. Refer to Procedure 008-006 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-008-006-om.html>)

Use additional supplemental coolant additive (SCA) to bring the coolant to the correct SCA concentration level. Refer to Procedure 018-004 in Section V.

(</qs3/pubsys2/xml/en/procedures/102/102-018-004.html>)



Install the pressure cap. Operate the engine until the coolant reaches a temperature of 70°C [160°F], and check for coolant leaks.



Fill

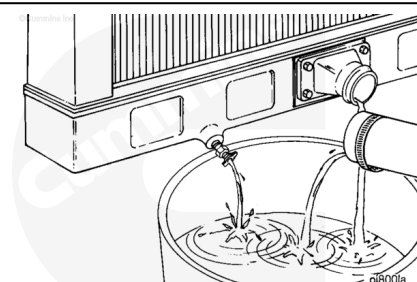
⚠ CAUTION ⚠

Use caution when filling the cooling system so that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations.

Close the radiator draincocks.

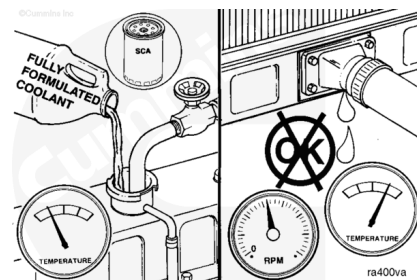
Install the lower radiator hose(s). Tighten the hose clamps.

Torque Value: 5 n·m [40 in-lb]

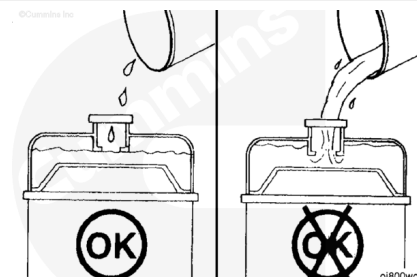


Use fully formulated coolant to fill the cooling system.

Use the correct units of SCA to obtain the correct cooling system protection.

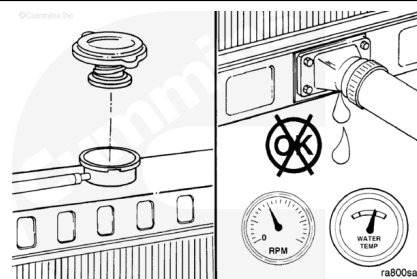


Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.



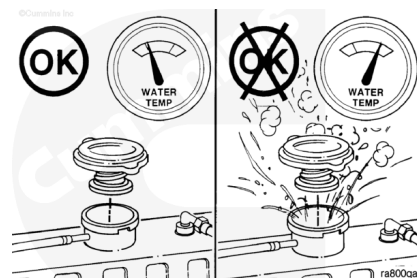
Replace the radiator or fill cap.

Operate the engine until the coolant reaches a temperature of 70°C [160°F]. Check for leaks.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Shut the engine OFF and allow it to cool.

Rail Applications

Refer to original equipment manufacturer (OEM) service manual.

